

Predictive Modeling & Risk Scoring for Bank Customer Churn

An Enterprise Early-Warning Machine Learning Framework for Retail Banking

Aditya Singh | Data Analytics & Quantitative Risk Track

Abstract

Customer churn directly degrades Customer Lifetime Value (CLV) and financial performance in retail banking. This paper introduces a quantitative machine learning risk-scoring system trained on 10,000 customer records. Incorporating engineered engagement ratios, product density indicators, and demographic profiles, our ensemble model yields robust predictive performance (AUC-ROC > 0.86), allowing financial institutions to transition from reactive churn analysis to proactive retention campaigns.

1. Analytical Methodology & Feature Engineering

Data Transformation: Removal of non-informative identifiers (CustomerId, Surname) followed by One-Hot Encoding on categorical attributes. Formulated Balance-to-Salary Ratio, Product Density Indicator, and Interaction Engagement Scores.

2. Model Evaluation Summary

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	79.2%	58.1%	28.4%	38.1%	0.762
Decision Tree	78.4%	48.2%	50.1%	49.1%	0.684
Random Forest	86.1%	74.5%	52.3%	61.4%	0.852
XGBoost Classifier	86.8%	75.2%	56.1%	64.2%	0.869